

Modeling Global Temperature Variate Effects Over Time

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Outline

- 1 Context
- 2 Processs
- 3 Appendix

What are we doing?

- Want to understand the dependence between climate variables (temperature, precipitation) and geography/time
- Want to create a model to capture the relationship between Latitude, Elevation, and Time, on Temperature and Precipitation

Why are we doing this?

- Want to understand the evolution of the parameters of the model in the modern climate
- Want to establish a baseline relationship of geography/time on climate variables to apply to ancient data
 - Paleogene era: 66-23 Mya
 - Proxy data (fossils, deposition layers, etc.)
 - Data Sparsity
 - Makes modeling covariate-driven? mean structure extremely important
 - $Y(s, t) = x(s, t)\beta + \epsilon(s, t), \epsilon \sim N(0, \sigma^2 I)$
 - Can learn appropriate structure for $x\beta$ from modern data (best design matrix)
- Put me through my paces

How are we doing this?

- Using the ERA5 reanalysis dataset [1]
- ERA5:
- Reanalysis:

Data Processing

- ERA5 data comes in a .25x.25 Latitude grid, with daily averages. Reduced this to a 1x1 Latitude grid with Weekly averages
- Used ERA5 Geopotential data, converted to elevation



Model



Bibliography

- [1] Copernicus Climate Change Service (C3S), *ERA5: Fifth generation of ECMWF atmospheric reanalyses of the global climate*, Climate Data Store (CDS), 2017. [Online]. Available: <https://cds.climate.copernicus.eu/cdsapp%5C#!/home>.